

From driven flow to inherited growth

Constructions and establishment tests

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Abstract

Can variation in a driven physical system become inherited differential growth? A complete answer requires a physical copying mechanism as well as a criterion for its spread. We separate channel formation, flow sorting, persistent patterns, reproduction with inherited differences, and portable information. Existing lattice and reaction–diffusion constructions establish some of these capabilities under specified rules; they do not yet realize the full sequence in one substrate. We then derive a conditional establishment test from measured usable power, copy cost, inheritance fidelity and the lifetime of copying ability. For a parent that accumulates work before each attempt, the critical cost is $E_c = W \log(1 + p)/\mu$. Replacing that clock by Poisson attempts with the same immortal-parent mean rate gives $E_c = Wp/\mu$, and can reverse the predicted outcome. Independent life-history simulations check the difference. Both results apply established branching and renewal mathematics. Their role is to make the proposed physical connection testable. A common microscopic construction of variation, copying and inheritance remains to be established.

1 The claim to establish

The long-term aim is to explain how local physical dynamics in an open, driven system can produce inherited differences that affect reproduction. The present claim is narrower: the necessary connections can be stated separately, some have constructive examples, and copying-clock assumptions have measurable consequences for establishment. A physical derivation must eventually join the connections under the same interaction rules. An analogy between replication and an epidemic does not establish that derivation.

The system exchanges matter or energy with maintained reservoirs. Let J_i be an intake throughput and W_i its usable power after conversion losses. For a specified driving potential, write $W_i = \epsilon_i J_i$, where ϵ_i is work per unit throughput, not necessarily a dimensionless efficiency. A work budget must also satisfy $\sum_i W_i \leq P_{\text{available}}$. Sustained currents require external supply; the argument assumes no maximum-entropy-production principle. The earlier concentration model has reversible coarse state dynamics despite representing a maintained throughput. It is therefore not by itself a resolved thermodynamic model of the driven apparatus.

2 Five separate physical questions

Question	What would establish it?
Paths form	Geometry develops under local material and transport rules.
Flow sorts	A measured small fraction of available paths carries a specified fraction of total throughput.
A pattern persists	Spatial organisation continues while its constituent material turns over.
A pattern reproduces	An identifiable parent produces a new structure, with perturbations that can persist across parent–offspring relations.
Information travels	A transported state recreates an inherited property at a separate site under common destination conditions.

These are not necessarily a chronological ladder. Unequal shares among fixed pipes establish sorting without circulation. A whirlpool can maintain its shape as water passes through it, yet need not preserve a particular perturbation or produce descendants. Spatially correlated patterns need not be related by copying. Inheritance of copying ability also has a lifetime distinct from the lifetime of the visible object. These distinctions determine which measurements matter.

The companion paper derives an exact stationary sorting window for a specified conserved-flow diffusion [1]. Independent contacts proportional to actual flow recover its variance-proportional-to-flow case. Deterministic lattice examples also seed unequal sharing from identical coarse states, without random draws after initialization. Deterministic effective noise has mathematical precedents [2]; the existing lattice’s correlations and limiting clock have not been fully matched to the theorem. This supplies a candidate source of variation, not yet inherited variation.

3 What the constructions establish

The restored lattice-Boltzmann pole model produces returning trajectories in its nearly steady wake, absent in low-drive, linear and no-pole controls. The hydrodynamic connection of suitable lattice-Boltzmann rules to Navier–Stokes is established [3]. A material extension replaces the fixed pole mask by binary sites with local cohesive exchanges. Their block-average occupancy sets fluid reflection. Three prepared patches retain about 96% spatial occupancy overlap and have 30–43 returning tracers out of 120 in the frozen final field. Equal material dispersed through the domain and uncoupled-material controls have no returning tracers in that diagnostic. Removing cohesion spreads the patch but leaves some circulation. The substrate supplies support; the experiment does not derive a macroscopic patch from uniform noise, free suspension or erosion.

The repository also reproduces known Gray–Scott spot multiplication from a finite seed using local reaction and diffusion laws [4, 5]. It contains no instruction to detect and copy a spot. The reaction network itself supplies autocatalysis; related chemical spot replication has been observed experimentally [6]. Splitting therefore has a constructive precedent, but these runs do not establish selectable inherited differences. Molecular templating provides another precedent for portable information simpler than a complete genetic system [7]; it has not emerged from the lattice models here. Existing token-carrier simulations instead assume copying and expression.

A prepared rare configuration can test a possible mechanism without waiting for its historical occurrence. It cannot measure its origin rate. To show that disturbance can create the capability, specify an initial ensemble and an allowed perturbation history that reaches

it under the same rules. A prepared capable state can be the endpoint of that construction, but its preparation alone does not establish reachability from the proposed background. The origin, establishment and spread of a capability remain separate questions.

4 From intake to establishment

Consider a rare lineage against a fixed background. A copying-capable structure receives constant usable power $W > 0$ and starts with an empty store. Every attempt costs $E > 0$, resets that store to zero and produces one viable daughter carrying the same copying capability with independent probability $p \in (0, 1]$. The parent survives a copying attempt. The daughter starts with an empty store. Copying ability is lost irreversibly at exponential rate $\mu > 0$; this may include physical death and loss of the inherited state. Copies are independent once born. Failed attempts still consume their work. Constant intake and independent descendants are rare-lineage approximations, not a model of population takeover under a fixed total budget.

Proposition 1. *For this accumulator model, positive probability of eventual lineage survival occurs precisely when*

$$\mathcal{R} = \frac{p}{\exp(\mu E/W) - 1} > 1, \quad E < E_c = \frac{W}{\mu} \log(1 + p). \quad (1)$$

Proof. Attempts occur at ages nt_c , $n = 1, 2, \dots$, where $t_c = E/W$. Survival of copying ability to age nt_c has probability $\exp(-\mu nt_c)$. The expected number of viable daughters over a parent's life is therefore $p \sum_{n \geq 1} \exp(-\mu nt_c)$, giving Eq. (1). Identical independent reproduction histories define a branching genealogy with nondegenerate offspring law. Its survival threshold is mean offspring greater than one [8]. Solving that inequality gives E_c . Birth ages affect growth in physical time; they do not change this genealogical criterion. $\square$

This is an application of established renewal and branching mathematics, not a new general threshold theorem. More generally the mean offspring is the integral of survival-weighted viable reproduction over age. Age-dependent reproduction cannot in general be replaced by a single mean attempt rate [9]. For immediate offspring in this particular accumulator model, the renewal equation also gives the mean-growth exponent $\lambda = W \log(1 + p)/E - \mu$. No claim about empirical growth or eventual population size follows without testing the model.

A parent with Poisson attempts at rate W/E has the same mean attempt rate if immortal, but a different lifetime output:

$$\mathcal{R}_P = \frac{Wp}{\mu E}, \quad E_{c,P} = \frac{Wp}{\mu}. \quad (2)$$

Because $\log(1 + p) < p$, the Poisson replacement can predict establishment where the accumulator predicts extinction. It permits attempts before enough work has arrived in an empty store, so it is not an equivalent pathwise resource model. At faithful copying ($p = 1$), the two critical costs differ by a factor $\log 2$. This comparison identifies a model substitution that needs justification.

For example, take $W = \mu = 1$, $p = .8$ and $E = .7$. The accumulator has $\mathcal{R} = .789 < 1$; the Poisson model has $\mathcal{R}_P = 1.143 > 1$. Thus a plausible change in the copying clock reverses the predicted outcome without changing the nominal attempt rate. A visible structure can also be long-lived while its copying state is short-lived: if these are independent irreversible losses with rates d and r , then $\mu = d + r$, not d alone.

Portable copying can enter through the same quantities. If transit independently destroys the required state at rate κ over duration τ , the surviving fidelity becomes $p = p_0 \exp(-\kappa\tau)$. Substitution in Eq. (1) gives a conditional transit-dependent critical cost. Transit also delays

births and changes the growth exponent; the immediate-offspring expression for λ then no longer applies. The transport law and the copying mechanism must be measured or constructed rather than inferred from a successful threshold fit.

For multiple inherited states with Markov transitions T , disappearance rates d , exponential attempt rates b , and viable daughter-state matrix P , the corresponding known criterion uses $K = (\text{diag } d - T)^{-1} \text{diag}(b)P$ and spectral radius $\rho(K) > 1$, under finite irreducible branching assumptions [8]. The repository checks this conditional model independently. Its rate-only state space cannot represent an empty-store accumulator without adding age or stored work. The scalar example shows why that distinction can change the conclusion.

5 Numerical checks and physical tests

Independent event histories check the lifetime output in Eqs. (1) and (2). Each setting uses 100,000 parents; a sampled loss time terminates explicit regular or exponential attempt times. Bernoulli outcomes record viable daughters. These are individual reproduction histories, not physical observations or complete population simulations.

Clock	Cost	Predicted daughters	Observed mean (SE)
Accumulator	.45	1.4077	1.4006 (.0058)
Accumulator	.70	.7891	.7836 (.0037)
Accumulator	1.00	.4656	.4643 (.0026)
Poisson	.45	1.7778	1.7643 (.0070)
Poisson	.70	1.1429	1.1355 (.0049)
Poisson	1.00	.8000	.7960 (.0038)

All six means lie within two estimated standard errors of their predictions; this is a descriptive numerical check, not a simultaneous confidence statement. The accumulator never spends work before receipt in the retained runs. Three unit tests check the survival-weighted sum, the classification reversal and work accounting. Raw outcomes, seeds and source hashes remain in the repository. Existing spatial, carrier and multitype checks are retained separately.

A discriminating experiment would measure usable intake, loss of copying ability, copy cost and parent–daughter fidelity independently of lineage outcomes. Time-resolved attempts distinguish accumulation from an exponential clock. Varying intake or cost then tests the predicted crossing of one viable daughter per lifetime; replicate genealogies assess establishment separately. Individual history and perturbation tracking must distinguish inheritance from common environmental forcing. More total copies are not sufficient if they systematically lose the copying state. Fixed-background predictions apply only while the lineage is rare; continued growth requires explicit accounting for competition and losses.

The strongest next construction would let a flow-maintained pattern acquire a copying capability through a finite perturbation, then transmit a modified property to descendants under unchanged local interaction rules. Parent and recipient manipulations should separate copied state from shared surroundings. A transported carrier would need to recreate that property at a new site. Parameters measured in those histories could then predict lineage outcomes on held-out runs. This would join physical possibility to a quantitative test; programming the copying operation would leave the central connection assumed.

6 Scope of the proposed contribution

The ambition is a physical account of inherited differential growth in driven systems. The constructions support individual capabilities; the copying-clock comparison states one conditional

prediction in independently measurable terms. Together they do not yet establish a common microscopic origin of variation, copying and portable inheritance. The decisive advance would be a construction that joins those capabilities under the same rules and predicts lineage outcomes from separately measured histories, or a physical test that distinguishes the proposed mechanism from alternatives. The present result gives that programme a concrete test while keeping its assumptions separate from its ambition.

Reproducibility. Sources, retained outcomes, assumptions and test commands are at <https://github.com/divadwg/selection-as-physics>, with the two-paper map in `paper/README.md`. The companion sorting paper is independent of this paper’s copying assumptions.

References

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